



SIMATS ENGINEERING

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**CSA1012-SOFTWARE ENGINEERING
CASE STUDY REPORT**

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Case Study:

SafeRoute AI: Intelligent Road Hazard Detection and Accident Prevention System

1. Problem Statement

1.1 Problem Explanation

Road traffic accidents are one of the major causes of injuries and deaths worldwide. Many accidents happen because drivers are unaware of hazards such as potholes, fallen trees, road debris, flooded roads, damaged traffic signs, poor visibility, and other unexpected obstacles. Current navigation applications mainly provide the shortest or fastest route but cannot detect these hazards in real time.

The proposed SafeRoute AI system uses Artificial Intelligence (AI), Computer Vision, GPS, IoT sensors, and predictive analytics to continuously monitor road conditions, detect hazards instantly, warn nearby drivers, recommend safer routes, and provide transportation authorities with real-time information for maintenance planning.

1.2 Objectives of the Proposed System

The objectives of SafeRoute AI are:

1. Develop an AI-powered road hazard detection system.
2. Detect potholes, debris, flooded roads, damaged signs, and other hazards using computer vision.

3. Integrate GPS for location-based hazard identification.
4. Send instant alerts and recommend safer alternative routes.
5. Predict accident-prone zones using historical and real-time traffic data.
6. Generate maintenance reports for transportation authorities.
7. Integrate IoT sensors and vehicle-mounted cameras for continuous monitoring.
8. Improve road safety and reduce accidents.

1.3 Scope of the Problem

The system covers:

- Real-time road hazard detection.
- GPS-based hazard mapping.
- AI-based image recognition.
- Driver notification system.
- Alternative route recommendation.
- Accident prediction using machine learning.
- Road maintenance reporting.
- Centralized hazard database.
- Smart city transportation integration.

Out of Scope

- Autonomous vehicle control.
- Vehicle insurance processing.
- Traffic law enforcement.
- Vehicle mechanical fault diagnosis.

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders

Stakeholder	Role
Drivers	Receive hazard alerts and safer routes
Transportation Authorities	Monitor hazards and schedule maintenance
Emergency Services	Respond to accidents quickly
Government	Improve road safety policies
Road Maintenance Teams	Repair damaged roads
AI Developers	Develop and maintain AI models
IoT Device Providers	Supply sensors and hardware
Navigation Service Providers	Integrate route guidance
Vehicle Manufacturers	Install cameras and sensors
Citizens	Report hazards through mobile app

2.2 Roles and Responsibilities

Drivers

- Follow hazard alerts
- Report road hazards
- Choose safer routes

Transportation Authorities

- Monitor hazard dashboard
- Prioritize road repairs
- Analyze reports

Emergency Responders

- Receive accident notifications
- Reach accident locations quickly

Government

- Allocate budgets
- Improve transportation policies

AI Engineers

- Train AI models
- Improve detection accuracy

IoT Engineers

- Install and maintain sensors
- Ensure continuous data collection

Maintenance Teams

- Repair roads
- Update repair status

2.3 Challenges Faced

Drivers

- Lack of real-time hazard information
- Unexpected road conditions
- Delayed warning

Authorities

- Slow manual reporting
- Difficulty prioritizing repairs
- Incomplete road condition data

Emergency Services

- Delayed accident detection
- Traffic congestion

AI Developers

- Large training datasets
- Weather affecting image quality
- False detections

IoT Providers

- Sensor failures
- Communication delays

3. Analyze the Current Approach

3.1 Existing System

Current systems include:

- Google Maps
- Waze
- Manual complaint portals
- CCTV monitoring
- Traffic police reporting

These systems mainly provide:

- Route navigation
- Traffic congestion updates
- Limited user-reported incidents

They cannot continuously detect road hazards automatically.

3.2 Strengths

- Widely available
- Easy navigation
- GPS tracking
- Large user base
- Traffic estimation

3.3 Limitations

- No automatic pothole detection.
- No AI-based obstacle recognition.
- Depends on manual reporting.
- Delayed hazard updates.
- No predictive accident analysis.
- Limited maintenance planning support.

3.4 Problems and Gaps

- Lack of continuous monitoring.
- Slow reporting process.
- No centralized hazard database.
- No real-time AI analysis.
- No predictive maintenance.
- Insufficient driver warnings.

4. Proposed Solution Using Software Engineering Principles

4.1 Proposed Software Solution

SafeRoute AI is a cloud-based intelligent transportation platform consisting of:

- Mobile Application

- AI Detection Engine
- GPS Module
- IoT Sensor Network
- Cloud Database
- Analytics Dashboard
- Maintenance Portal
- Emergency Alert Module

The system continuously collects road images from vehicle-mounted cameras and IoT sensors. AI analyzes the images, identifies hazards, stores the location in the cloud, and immediately alerts nearby drivers.

Transportation authorities receive maintenance requests through an administrative dashboard.

4.2 Functional Modules

Module 1: User Authentication

- Login
- Registration
- Role management

Module 2: AI Hazard Detection

- Image processing
- Object detection

- Hazard classification

Module 3: GPS Mapping

- Location tracking
- Hazard mapping
- Navigation integration

Module 4: Driver Alert System

- Voice alerts
- Push notifications
- Alternative route suggestions

Module 5: Predictive Analytics

- Accident prediction
- Risk score generation
- Traffic pattern analysis

Module 6: Authority Dashboard

- Hazard reports
- Maintenance requests
- Analytics visualization

Module 7: Emergency Response

- Accident notification
- Location sharing
- Emergency contact alerts

Module 8: Database Management

- Hazard storage
- Historical data
- User reports

4.3 Functional Requirements

- Detect hazards automatically.
- Capture camera images.
- Identify potholes and obstacles.
- Store GPS coordinates.
- Notify nearby drivers.
- Recommend alternate routes.
- Generate maintenance reports.
- Predict accident-prone areas.
- Maintain historical database.
- Support administrator dashboard.

4.4 Non-Functional Requirements

Performance

- Detection within 2 seconds.

Reliability

- 99.9% uptime.

Scalability

- Support millions of users.

Security

- Encrypted communication.
- Secure authentication.

Availability

- 24×7 service.

Usability

- Simple mobile interface.

Maintainability

- Modular architecture.
- Easy software updates.

4.5 Software Engineering Principles

Modularity

Separate modules for detection, GPS, alerts, analytics, and reporting.

Scalability

Cloud infrastructure allows handling increasing users and sensors.

Maintainability

Independent modules simplify updates and debugging.

Reliability

Redundant cloud servers and backup databases improve reliability.

Usability

User-friendly interface with minimal driver distraction.

Security

- Authentication
- Authorization
- Data encryption
- Secure APIs

5. Justification of SDLC Methodology

Selected Methodology: Agile Scrum

Why Agile Scrum?

SafeRoute AI requires:

- Frequent AI improvements

- Continuous testing
- Customer feedback
- Feature updates
- Sensor integration

Agile supports all these requirements.

Advantages

- Incremental development
- Faster delivery
- Easy requirement changes
- Continuous testing
- Better stakeholder communication
- Frequent software releases

Agile Sprint Plan

Sprint 1

- User authentication
- Database setup

Sprint 2

- AI hazard detection

Sprint 3

- GPS integration

Sprint 4

- Driver notification

Sprint 5

- Analytics dashboard

Sprint 6

- Testing and deployment

How Agile Supports Maintenance

- Regular bug fixes
- AI model retraining
- Feature enhancement
- Continuous deployment
- Easy scalability

6. Expected Outcomes and Limitations

6.1 Expected Benefits

- Early hazard detection.
- Reduced accidents.
- Lower vehicle repair costs.
- Faster emergency response.
- Better road maintenance planning.

- Accurate hazard mapping.
- Improved traffic management.
- Support for smart city development.

6.2 How the Solution Addresses Challenges

Challenge	Solution
Delayed hazard reporting	Real-time AI detection
Poor road visibility	Computer vision analysis
Manual maintenance requests	Automatic reporting
Unsafe navigation	Instant driver alerts
Accident prediction	Machine learning analytics
Maintenance prioritization	Authority dashboard

6.3 Risks and Limitations

Risks

- Poor internet connectivity.
- Camera obstruction.
- Sensor malfunction.
- Incorrect AI predictions.
- Privacy concerns.

Limitations

- Requires camera-equipped vehicles.
- High initial deployment cost.
- Performance affected by severe weather.
- Continuous AI training required.
- Dependence on cloud connectivity.

6.4 Future Enhancements

- 5G-enabled communication.
- Drone-based road inspection.
- Vehicle-to-Vehicle (V2V) communication.
- Vehicle-to-Infrastructure (V2I) integration.
- Deep learning model improvements.
- Smart traffic signal integration.
- Autonomous vehicle compatibility.
- Edge AI for faster hazard detection.
- Voice assistant support.
- Integration with emergency healthcare systems.

Conclusion

SafeRoute AI provides an intelligent and scalable solution for improving road safety through AI, computer vision, GPS, IoT,

and predictive analytics. By automatically detecting road hazards, providing instant driver alerts, predicting accident-prone areas, and assisting transportation authorities with maintenance planning, the system addresses the limitations of existing navigation platforms. Using the Agile Scrum methodology ensures continuous improvement, rapid feature delivery, and easier maintenance. Overall, SafeRoute AI supports safer roads, efficient traffic management, and future smart city transportation initiatives.

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